



GAYU BARUWA

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SUMMARY

Informatics Engineering student at ITS Surabaya with a primary focus on Machine Learning, NLP, and Data Science, complemented by full-stack web development experience (internship at GMF AeroAsia, Laravel/Next.js projects). Completed Dicoding's Machine Learning Bootcamp (Batch 8).

EXPERIENCE

PT Garuda Maintenance Facility AeroAsia Tbk (GMF AeroAsia)

July 2025 - October 2025

Internship – Web/App Development

- Developed internal business applications using Microsoft Power Apps, with SharePoint Lists as the data backend
- Automated operational workflows using Power Automate
- Collaborated with cross-functional teams to translate business requirements into technical solutions

Dicoding Indonesia

June 2025 - November 2025

Dicoding Bootcamp Batch 8 – Machine Learning

- Completed end-to-end ML pipeline: data preprocessing, feature engineering, model training, evaluation
- Built supervised and unsupervised learning models on real-world datasets

PROJECTS

Face Recognition with ArcFace and ResNet50 (Group Project)

- Implemented ArcFace loss (Deng et al., IEEE T-PAMI 2021) from scratch in PyTorch, using a pretrained ResNet50 backbone with a 512-dim BN-Dropout-FC-BN embedding head, following the paper's exact hyperparameters ($s=64$, $m=0.5$, SGD with MultiStep LR)
- Trained on CASIA face dataset (200 identity classes, ~20K images, 80/20 split) on Google Colab with a Tesla T4 GPU over 32 epochs
- Achieved 50.08% Top-1 accuracy (100× above random chance), 66.07% Top-3, and macro-average AUC of 0.9557 across all 200 classes, evaluated with confusion matrix, ROC curves, and t-SNE embedding visualization

Skincare Compatibility Checker (Capstone Project)

- Cleaned, deduplicated, and merged skincare datasets from 3 sources (Kaggle Cosmetics, Sephora Ingredients, product catalogs) into a unified corpus of 4,082 products spanning 116 brands and 6,696 unique INCI ingredients, normalized into 13 functional categories
- Built a rule-based NLP/ML engine to detect skincare ingredient incompatibilities, covering 6 major active-ingredient groups (retinoids, vitamin C, AHA, BHA, niacinamide, benzoyl peroxide) across 7 pairwise interaction rules with graded AVOID/CAUTION severity levels
- Developed a content-based recommendation engine using cosine similarity over ingredient-function profiles ($4,082 \times 4,082$ similarity matrix), integrated both models into a FastAPI backend and collaborated with Flutter developers on mobile deployment

EDUCATION

Bachelor of Informatics Engineering

2022 – Present

Institut Teknologi Sepuluh Nopember, Surabaya

- GPA: 3.41/4.00

SKILLS

Machine Learning & Data

- Data preprocessing and exploratory data analysis (EDA)
- Supervised learning (classification, regression)
- Unsupervised learning (clustering)
- Text representation and embedding
- Model evaluation and performance metrics
- Feature engineering and dimensionality reduction

Web & Software Development

- JavaScript, PHP, HTML/CSS, Laravel, React/Next.js
- Git & GitHub
- Microsoft Power Platform (Power Apps, Power Automate, SharePoint)

ORGANIZATIONAL EXPERIENCE

Himpunan Mahasiswa Teknik Computer - Informatika ITS (HMTC)

March 2024 - December 2024

Staff – Student Talent and Interest

- Coordinated weekly training schedules for the Informatics Futsal Club.
- Organized friendly Futsal matches for all Informatics Engineering students.

Schematics REEVA

February 2024 - November 2024

Expert Staff – Festival Division

- Developed storyline and scripts for the REEVA 2024 musical drama performance.
- Served as Front of House (FOH), acting as the primary controller for sound system, lighting, and multimedia during live events.

UKM Golf ITS

February 2024 – December 2024

Head of Finance Division

- Managed budgeting and procurement for merchandise production.
- Acted as liaison for internal logistics and financial coordination.